

The Standard Model as a decompactified projection

FFGFT as the unifying explanatory model – why every calculation with α , v or Yukawa carries the recursion along

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Contents

Doc. 298 The Standard Model as a decompactified projection — FFGFT as the unifying explanatory model

Starting point: the constants used are not neutral inputs

Every calculation that uses the fine-structure constant α , the Higgs vacuum expectation value v , or a Yukawa coupling carries the K_{frak} recursion *indirectly* with it, from the FFGFT point of view — whether or not this is named. For these quantities are not free inputs in FFGFT but outputs of the recursion:

- $\alpha = \xi E_0^2$ (Doc. 006/292), where E_0 already contains the K_{frak} factor ($1 - 100\xi$) absorbed ($7,398/7,348 = K_{\text{frak}}^{-1/2}$, Doc. 190/P43). Whoever uses α uses a value into which the recursion has entered.
- The masses are the ξ^p ladder ($m = r\xi^p v$, Doc. 292); a Yukawa coupling $g = m/v$ is therefore already a ξ -ladder ratio. Whoever writes Yukawa writes a recursion ratio in different notation.

The Standard Model takes these quantities as *measured* inputs; FFGFT *derives* them. Therein lies the structural relationship this document describes.

The decompactified domain

Whoever uses α , v , or Yukawa couplings in continuous fields and potentials on $\mathbb{R}$ — a Yukawa potential $V(r) \propto e^{-mr}/r$ on open space, a continuous scalar field, a mediator scale — works structurally in the *unrolled* picture. This is exactly the domain into which FFGFT decompactifies the compact recursion on T^4 (type-I decompactification, Doc. 270): the compact origin of the constants is carried along but not named.

This is the same pattern as reduction without a named whole from Doc. 296/297: a projection that evaluates correctly but does not name the object of which it is a segment. With U_3 (Doc. 297) it was the time window without the full winding; here it is the decompactified constant sector without the compact recursion from which α and the Yukawa ratios originate.

FFGFT as the unifying explanatory model

Here lies the actual, positive content. FFGFT connects quantum mechanics and the Standard Model not by replacing them but by placing both as *projections of the same compact object* (Doc. 296): QM and relativity are projections onto different scales, and the Standard Model is the decompactified single-scale projection of its constants.

From this follows a property that distinguishes FFGFT among the known frameworks: it is in principle able to *incorporate any already-working partial model as a projection*. A model that computes correctly on its scale — the Standard Model in the particle sector, atomic physics in the orbital sector, an effective scalar channel in precision spectroscopy — is, from the FFGFT view, not a competitor but a projection for which FFGFT specifies the place where it sits. The working partial model is not refuted; it is located.

The mechanism for this is reduction as the condition of evaluability (Doc. 296): every projection leaves out and thereby makes a partial domain evaluable. FFGFT supplies the whole of which these partial models are segments — and can therefore incorporate them, provided they work on their scale. What a partial model correctly describes on its scale remains correct; FFGFT adds from which compact object the constants used originate and how the scale connects to the others.

The placement is conditional (P35)

Note on referencing: "P35" denotes, throughout the corpus, the correction discipline maintained in the correction register of Dok. 190 (there as ξ^N triviality: a quantity of the form $X = \xi^N$ or $X = \varphi^N$ is by itself void of content; only a forward-derived exponent counts as content; "candidate" $\neq$ "derived", verify numbers before claiming). The uses of P35 in this document are to be read in this sense.

The statement "the Standard Model is a decompactified subset of FFGFT" is an *FFGFT-internal* placement, not an externally enforceable proof. From the Standard Model's point of view, α and v are measured natural constants — full stop. The subset relation holds *under the assumption* that FFGFT's derivations are correct: *if* $\alpha = \xi E_0^2$ and the ξ^p mass ladder are correct, *then* the Standard Model is the unrolled projection that uses these constants as inputs without naming their compact whole.

This conditionality is to be maintained — the same discipline FFGFT also demands of competing cosmological models. The Standard Model can neither refute the containment relation (it computes correctly in the unrolled picture) nor confirm it from within (it names no compact object). It is a placement, not a claim about what the Standard Model itself asserts.

What is established and what remains programme

To keep separate:

- **Established** is the containment relation for the sector FFGFT actually derives: α (Doc. 006/292), the lepton masses and the Koide scalar $Q = 2/3$, the phase $\theta = 2/9$ (Doc. 291/293). For these quantities it is shown that they are recursion outputs.
- **Programme** remains the *full* Yukawa matrix: quark masses, mixing angles (CKM/PMNS), CP phase. For these the subset relation is not worked out — here "the Standard Model is a subset" would be a promise, not a shown derivation. This part is to be carried as an open obligation, not as a result.

This separation is the same as throughout the corpus: core derivations (proved from ξ), bridges (algebraically proved), reductions (plausibility sketches). The SM subset relation is core derivation for the lepton sector, programme for the full Yukawa matrix.

Extension within the projection — the history of the Lagrangian

A misunderstanding must be cleared up here. The question is not whether the Standard Model is "complete" — that is too broad. The Standard Model *already is* a projection onto a partial domain: the decompactified single-scale projection of its constants. Any attempt to extend it therefore takes place *within this already-projected partial domain*.

This has a structural consequence that holds before any concrete extension: in every projection one can *refine*, but not go *beyond* its edge. A projection has already left out the compact whole; an extension within the projection cannot retrieve what was left out. It can only place more structure into the already-projected segment. The edge of the projection is the limit of the extension — and this limit stands fixed before the extension begins.

This is exactly the history of the Standard Model and its Lagrangian. One sees an incompleteness — neutrinos are massive after all, an anomaly appears, a sector does not fit — and *patches on*: a further term, a further field, a further effective coupling is added to the Lagrangian. The Lagrangian grows, but it grows *within the same projection*. Every addition stays in the decompactified partial domain; none crosses its edge towards the compact whole. Patching refines the projection, it does not switch it.

The telling evidence for this is that every such extension must *import* its building blocks. The quantities an extension needs — α , v , mass ratios — are not derivable within the partial domain, because they belong to the left-out whole. They are brought in from outside, as measured inputs. That an extension must import, rather than generate from its own framework, is the sure sign that it works within the reduction: what is missing belongs to the whole that the projection does not name.

The difference from FFGFT is therefore not gradual but structural. FFGFT does not patch the projection — it names the compact whole from which the projection and its constants originate. It does not extend within the partial domain but gives the partial domain as a segment of a whole. This is why FFGFT need not import α : it derives α from ξ (Doc. 006/292). The import is the signature of extension within the projection; the derivation is the signature of the named whole.

The inner view — a pre-Copernican analogy

Patching within the projection has a historical precedent. Pre-Copernican astronomy was not *wrong* in the sense of poor calculation — epicycle astronomy predicted planetary positions precisely for centuries. It was an *inner view*: the perspective of an observer who takes their own location to be the centre. When a deviation appeared, one patched on — a further epicycle, a shifted deferent, an equant. The system grew but stayed within the same inner perspective.

This is structurally the same pattern: the inner view = the projection that takes its own location (the one scale) to be the whole; patching on epicycles = adding terms to the Lagrangian; precise but not explanatory = the calculation holds on the scale but does not name the whole.

The Copernican step was not a better fit — at first it was even worse (Copernicus kept circular orbits). It was a *change of perspective*: the system named as a whole from which the apparent motions follow as projections. The retrograde motion of the planets was no longer an epicycle but a projection effect of the Earth's motion — the same phenomenon, but located rather than patched on.

This is exactly how FFGFT relates to the Standard Model: the apparent arbitrariness of the 19 parameters is, from the inner view, a set of numbers to be adjusted; from the outer view they are projection effects of a recursion ($\alpha = \xi E_0^2$, the ξ^p mass ladder). Not fitted better — located differently.

Two limits keep the analogy honest (P35). *First*, it describes the *structure* of the two views but does not guarantee the historical outcome: Copernicus prevailed because Kepler and Newton made the outer view *quantitatively* superior. FFGFT is conceptually simpler (one parameter instead of nineteen) and derives the lepton sector; full quantitative superiority across all sectors remains programme. *Second*, the very fact of unlimited patchability is the mark of the inner view: it can never fail from within, only be overtaken from without. That the Standard Model can absorb every new anomaly through a further EFT term is therefore not a sign of its strength but the signature of an inner perspective.

HLV-YUKAWA-BRIDGE1 as a special case

Marcel Krueger's HLV-YUKAWA-BRIDGE1 (Zenodo, DOI 10.5281/zenodo.21224030) makes this pattern explicitly visible. His effective bridge strength is parametrised as

$$g_\chi^2 = \eta_\chi \alpha \varphi_g^{-2k}, \quad \varphi_g = \frac{1 + \sqrt{5}}{2}, \quad k \in \mathbb{N}. \quad (1)$$

This construction stands with one foot in FFGFT territory and one in HLV territory, in the unrolled space: α is an FFGFT recursion output (which carries the recursion along, see

above), φ_g is the HLV-side eigenvalue that the same crystallographic restriction ($2 \cos(2\pi/5) = 1/\varphi \notin \mathbb{Z}$) forbids in FFGFT and assigns to HLV (Doc. 190/R44). Marcel himself names η_χ as the main vulnerability and warns that the construction could, without leakage-free fixing of k and η_χ , degenerate into parametrisation rather than prediction — the same P35 discipline.

Thus this more conservative HLV channel too is structurally *downstream* of FFGFT: a decompactified construction using an FFGFT recursion output (α) as a building block. It cannot claim to be independent of the recursion content of that quantity — it is carried along, whether named or not. The sector itself (isotope-shift/King-plot, nucleus–electron scalar force) is disjoint from the FFGFT lepton sector; there is no conflict but a placement relation: a working partial channel that FFGFT can in principle incorporate as a projection.

What Marcel would have to prove

For the Yukawa bridge to become a falsifiable channel — and further, to step out of the projection — a staged proof burden must be carried. The first four points are his stated programme and methodologically the right path; the fifth is not laid out within his framework.

Within the projection (falsifiable channel):

1. **Fix η_χ independently.** As long as η_χ is a free adjustment factor, Eq. (9) is a parametrisation, not a prediction. η_χ would have to be *derived* from HLV geometry, not fitted to data. Marcel names this himself as the main vulnerability — the test is not yet passed.
2. **Lock k before the holdout data.** The index $k \in \mathbb{N}$ must be fixed before the King-plot data are examined — otherwise target leakage. Pre-registration discipline, necessary, but not a physics proof.
3. **Beat the generic scalar (his N_3 null).** If a generic scalar mediator with free coupling and mass explains the data as well as the φ_g^{-2k} -locked parametrisation, the HLV *specificity* is not supported. He would have to show that the φ structure hits something the free scalar does not. This is the hardest of the four.
4. **Survive the ordinary nonlinearities (N_1/N_2).** King-plot deviations also arise from nuclear structure (finite charge distribution, higher mass shifts) and many-body effects — entirely without a new scalar (his references on Yb, Ca). He would have to show that a signal remains *after* subtracting these known effects.

If he passes (1)–(4), he has a genuine, falsifiable BSM channel — a clean refinement of the partial domain. But he would thereby remain *within* the projection, because he imports α and φ .

Out of the projection (change of perspective):

5. **Derive η_χ and the φ power from a named whole, instead of importing them.** Only if it were shown where α and φ in the formula *come from* — from which compact structure — would the bridge move from patching to a change of perspective. As long as α is an imported input (from the FFGFT view a recursion output) and φ a *held*, not *derived*, eigenvalue (Doc. 190/R44), the bridge refines the inner view; it does not leave it.

The finding is thus cleanly separated: (1)–(4) are necessary for a falsifiable channel and are Marcel's stated programme; (5) is the step out of the projection and is not laid out within his framework — HLV *has* φ as an eigenvalue but does not derive it, and α is taken from outside. The import remains the signature of extension within the projection.

Status

FFGFT-internal placement, conditionally formulated (P35). Established for the lepton sector (α , Koide, $\theta = 2/9$ as recursion outputs), programme for the full Yukawa matrix. The positive content: FFGFT is the unifying explanatory model that places QM and the Standard Model

as projections of the same compact object and can in principle incorporate any partial model working on its scale as a projection — not as a refutation but as a placement. No unconditional claim about what these partial models assert from within.